Who Burns the Tokens? Fiscal Sustainability and Demand Concentration in GeoDePIN Networks

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Abstract

This paper analyzes a class of DePIN protocols the paper terms GeoDePIN: networks whose value depends on where hardware is deployed, not merely that it exists. Location-dependence creates design pressures absent from compute-only DePIN alternatives, and the paper evaluates how protocols respond using on-chain evidence from 13 networks and a 34-month longitudinal case study of Helium. Helium became the first GeoDePIN protocol where paid usage fully offsets token emissions, with its Subsidy-to-Revenue ratio (S2R) rising from 0.013 under coverage-based rewards to 1.84 between May 2023 and February 2026. Data Credit burns now exceed HNT emissions by approximately 84%, producing net annual deflation of 3 to 4%. However, roughly five purchasers account for 90% of those burns: a demand concentration that aggregate fiscal metrics cannot detect. DIMO (S2R 4.24) subsequently crossed fiscal parity; GEODNET (0.219) and Hivemapper (0.183) occupy a growth-phase middle tier; the remaining protocols cluster near zero. Governance concentration across the thirteen-protocol GeoDePIN cross-section examined here (Herfindahl-Hirschman Index 0.034 to 0.593) materially exceeds the post-exclusion DeFi benchmarks from a companion cross-sectional study (0.005 to 0.065). Four outputs organize the findings: a classification distinguishing GeoDePIN from location-independent protocols with testable predictions, a design playbook mapping mechanisms to protocol maturity stages, a health dashboard with demand-concentration guardrails, and a subtraction test for mechanism necessity.

Keywords: DePIN; GeoDePIN; Subsidy-to-Revenue ratio; burn-and-mint equilibrium; demand concentration; Helium; service verification; token velocity

JEL Codes: D43, D47, L14, L22, O33, R12

List of Abbreviations

VWAP Volume-Weighted Average Price

TWAP Time-Weighted Average Price

S2R Subsidy-to-Revenue ratio

RTK Real-Time Kinematic (satellite positioning)

HIP Helium Improvement Proposal

HHI Herfindahl-Hirschman Index

GeoDePIN Geographically Distributed DePIN

EIP Ethereum Improvement Proposal

DeVIN Decentralized Virtual Infrastructure Network

DePIN Decentralized Physical Infrastructure Network

DC Data Credits (Helium network)

1. INTRODUCTION

1.1 The Coordination Problem

In mid-2024, Helium was generating roughly $6,500 per month in paid usage across 975,000 IoT hotspots [60], [61]; revenue so thin that 99% of operator income came from token subsidies rather than customers. By Q3 2025, after a governance-led reform (HIP-147) shifted rewards from coverage to verified service, annualized paid usage had reached $18.3M [28] and token burns exceeded emissions. The same network, the same hardware, a different rule: this is the institutional design problem DePIN poses. What mechanisms distinguish protocols that cross fiscal parity from those that remain subsidy-dependent? DePIN systems coordinate physical infrastructure through token incentives [22]; when the design is wrong, the failure modes are coverage mining, subsidy dependence, and governance capture.

GeoDePIN protocols constitute the most demanding stress test for institutional design [13] in the token economy: verification must prove physical presence under adversarial conditions, hardware lock-in raises the stakes of governance failure, and subsidy taper creates fiscal cliffs that force demand-driven sustainability.

1.2 GeoDePIN vs. DeVIN: A Taxonomic Distinction

A shared coordination pattern distinguishes these protocols from other DePIN projects: their networks derive competitive advantage from geographic coverage density across unique physical locations. We term this subcategory GeoDePIN (geographically distributed decentralized physical infrastructure) to distinguish it from DeVIN (decentralized virtual infrastructure networks) such as Filecoin [15] (storage), Livepeer (video transcoding), or Render (GPU compute), where defensibility comes from compute density, throughput, or cost efficiency rather than location.

GeoDePIN projects offload capital expenditure to retail participants who bet that the network's future utility will justify their hardware investment. Four design pressures are distinctive to GeoDePIN: (1) Location is a first-class economic variable: the value of a Helium hotspot in downtown Manhattan differs from one in rural Wyoming by orders of magnitude. (2) Verification must prove physical presence, location authenticity, and service delivery under adversarial conditions, which is why Helium's HIP-147 transition to proof-of-service was decisive. (3) Subsidy taper threatens network contraction: when token rewards decline, operators in low-demand areas face negative unit economics first. (4) Incentives must be spatially intelligent: uniform global reward rates systematically over-reward saturated areas and under-reward frontier coverage.

The capability framework [48] interprets these design pressures as capability constraints: when operators absorb irrecoverable hardware sunk costs, their substantive freedom to exit the network is structurally negated, making governance voice mechanisms, not market exit, the primary channel for expressing dissatisfaction. This asymmetry is absent from DeVIN protocols, where compute resources can be redirected across networks without capital loss, and it motivates the bicameral governance design developed in Section 2.

Network coverage models predict that GeoDePIN protocols face location-dependent tipping points absent from digital resource networks: Soria Ruiz-Ogarrio et al. [39] formalize optimal coverage equilibria for DePIN deployments, while Corn et al. [40] demonstrate empirically that spatial density determines forecasting accuracy in solar DePIN networks, consistent with geographic saturation imposing diminishing returns specific to location-dependent protocols.

The GeoDePIN design space can be summarized as the intersection of three optimization problems: location economics (where to deploy), verification integrity (how to prove real service), and subsidy discipline (when to taper). Section 6 develops these into a playbook; the taxonomy is formalized in Section 4.

1.3 Research Questions

Two research questions guide the analysis:

RQ1: What institutional mechanisms distinguish sustainable GeoDePIN protocols from fragile ones? Specifically, how do sink architecture, reward structure, and governance design interact to determine whether a protocol achieves fiscal self-sufficiency?

RQ2: How should design playbooks differ for GeoDePIN (geographically distributed, retail-operated) versus DeVIN (compute-dense, institutionally operated) architectures? What testable predictions does this taxonomic distinction generate?

1.4 Paper Outline

Sections II through VII move from design architecture and methods through taxonomy, empirical evidence, and design playbook to conclusion. Practitioner and policy briefs appear as Supplementary Materials S1 and S2.

1.5 Positioning

DePIN has attracted growing attention from both industry and academic communities. Messari's annual DePIN reports [2], [28] track sector-level metrics (market capitalization, node counts, revenue) but do not evaluate institutional design quality. IoTeX's "DePIN sector map" classifies protocols by service type but does not distinguish architectural differences in operator base or governance structure. Borderless Capital has proposed a GeoDePIN framing emphasizing geographic coverage as a moat, but without the taxonomic precision or testable implications developed here.

In token engineering, Voshmgir [19] and the Token Engineering Commons have advanced design frameworks centered on bonding curves (mathematical functions that set token price as a function of supply) and agent-based simulation. These approaches optimize for mechanism efficiency but typically omit the normative constraints (legitimacy, fairness, contestability) that the companion framework paper argues are essential for institutional durability. The mechanism design [10], [11] literature (Hurwicz, Myerson) optimizes for incentive compatibility but does not address governance concentration or capability dimensions that DePIN uniquely raises when operators are retail individuals investing personal capital.

Lin et al. [53] provide an architectural survey formalizing the multi-layer DePIN design stack and its integration with burn-and-mint equilibrium mechanics; the present paper extends this engineering perspective by adding institutional design evaluation and longitudinal fiscal sustainability measurement.

The safety and reliability engineering community has begun analyzing DePINs as legitimate critical physical infrastructure paradigms, evaluating their resilience properties through established frameworks for distributed ownership and graceful extensibility [54].

This paper applies an institutional design lens to the specific coordination challenges of geographically distributed infrastructure, contributing a taxonomy, empirical evidence, and a practitioner-oriented playbook that existing DePIN analyses lack.

2. DESIGN ARCHITECTURE

The normative framework motivating these design choices draws on Kantian publicity, Rawlsian fairness, republican non-domination, and Hayekian knowledge-use. This paper retains Ostrom's polycentric governance framework [14] as directly relevant to the geographic heterogeneity that distinguishes DePIN from other token-coordinated systems. Recent frameworks have systematically mapped Ostrom's eight design principles to decentralized community governance [31]; the GeoDePIN taxonomy developed here extends this mapping by specifying how geographic heterogeneity creates governance challenges absent from purely digital commons.

DePIN requires an institutional architecture built around four interacting loops.

2.1 Design Objectives and Constraints

The mechanism design architecture implements the theoretical framework under DePIN conditions, treating protocol rules as economic institutions [26] that must satisfy four objectives simultaneously:

Economic sustainability: incentives must not require perpetual inflationary subsidy.

Service reliability: rewards must track verifiable service delivered, not merely presence.

Political legitimacy: rules must be publicly defensible; enforcement must be contestable.

Social impact: system success should include capability gains and inclusion, not only financial metrics.

Constraints include pseudonymity, sybil threats (one actor creating multiple fake identities to claim disproportionate rewards), geographic heterogeneity, oracle error (inaccurate or stale data from external data feeds), governance capture risk, and legal/regulatory uncertainty.

2.2 The Prime Directive: Sinks Before Faucets

The sinks-before-faucets principle motivates the following architectural constraint: A repeated fragility pattern across token systems is reward-first bootstrapping without durable demand coupling [20]. This paper formalizes the directive:

Principle 2.1 (Sinks Before Faucets): A system should deploy and verify at least one usage-linked sink before scaling emissions; burns per unit service, fee routing to treasuries that reduce circulating float.

Alshater [41] synthesizes the DePIN flywheel mechanism and the necessity of fiat-denominated pricing rails in a recent scoping review (Frontiers in Blockchain, 2026), identifying the prevailing "DePIN Flywheel" pattern grounded in a Burn-and-Mint Equilibrium "where demand is monetized through fiat-denominated usage credits created by burning the network's native token." Alshater further characterizes the architecture as a composition of three primitives: the burn-and-mint sink, governance-calibrated issuance, and collateral requirements that bind operator participation to network state. The review identifies four structural challenges that the longitudinal S2R analysis developed here addresses directly: token price volatility impact on provider economics, robust incentive alignment, generating sufficient non-speculative demand, and regulatory uncertainty. Alshater's research agenda explicitly prioritizes empirical event studies of governance changes and quality-adjusted reward measurement; the longitudinal S2R analysis developed here provides the empirical validation that survey identifies as missing from the current literature.

Violating this directive makes the system dependent on continued speculative inflows, a constitutional [4] fragility.

2.3 Dual-Asset Economics

DePIN systems face an adoption barrier: mainstream users prefer predictable stable pricing, while operators require upside and a credible long-run investment case. A dual-asset model, extending two-sided market logic [16], separates these needs:

Definition 2.1 (Dual-Asset Model). Let (T) be a governance/stake token and (C) be non-transferable usage credits priced in stable units [20], where:

(T) governs policy and secures operator performance (staking/slashing).

(C) is spent for service usage and obtained via (i) burning (T), or (ii) paying fiat/stables that triggers protocol buy-and-burn of (T) and mints (C).

Helium's Data Credits [2] provide an operational precedent for non-transferable usage credits spent on network usage and onboarding.

Institutional effect: users buy predictable service; the protocol translates demand into sink pressure; operators are compensated via governed reward policy.

2.4 DePIN Incentives: Pay for Service, Not Presence

DePIN's central risk is paying for "presence" (hardware online) rather than service delivered. When rewards track presence, operators deploy hardware in locations with no demand, diluting rewards for operators who serve real users and eroding network credibility [2].

Principle 2.2 (Reward Decomposition). Total operator rewards per epoch (the protocol-defined reward period, typically one day or one week) R_t should be decomposed as:

$$R_t = R_t^{base} + R_t^{usage}$$

R_t^{base} maintains baseline coverage and supports frontier bootstrapping.

R_t^{usage} dominates over time and pays only for SLA-verified service.

Definition 2.2 (Quality-Adjusted Usage Reward). Let $q_{i,t}$ be a quality score for operator (i) in epoch (t), derived from SLA indicators (uptime, latency, accuracy, integrity). Let $u_{i,t}$ be verified usage delivered (demand-weighted). Then:

$$R_{i,t}^{usage} \propto u_{i,t} \cdot q_{i,t}$$

subject to anti-gaming constraints and caps.

Proposed maturity schedule (normative default)

Early: 40% base / 60% usage

Mid: 25% base / 75% usage

Mature: 10% base / 90% usage

This schedule is a policy hypothesis rather than a universal claim; it is evaluated empirically via event studies and cohort comparisons.

2.5 Reflexivity-Aware Emissions: The S2R Autopilot

Token incentives are reflexive [18]: the token price affects the real cost of emissions and can induce boom-bust deployment cycles. This section proposes a rule-based stabilizer:

Principle 2.3 (Usage-Bounded Reward Budget). Reward budgets should be set in stable-value terms as a function of real usage revenue, not as an arbitrary token issuance number.

Rev_t be usage-derived revenue in epoch (t) (or equivalent proxy).

$Budget_t = \alpha \cdot Rev_t$ for some governed spending fraction (α).

$Price_t^{TWAP}$ be a smoothed price measure.

$$F_t = \frac{Budget_t}{Price_t^{TWAP}}$$

$$F_t \in \left[(1 - \delta)F_{t-1},\ (1 + \delta)F_{t-1}\right]$$

for (δ) (e.g., 5–10% per epoch). Monitoring variable: S2R

$$S2R_t = \frac{S_t}{F_t}$$

The policy aim is not a fixed number but a governance-selected band, with automatic responses when outside bounds.

Why this is institutional (not only technical): the rule constitutes a fiscal constitution [4]: an explicit commitment to disciplined spending and transparency, consistent with transparency and signal responsiveness principles. Proposed trigger conditions (Proposal, Normative)

These triggers extend the bounded adjustment logic above. The S2R trigger prevents runaway inflation when usage sinks fall below half of issuance, a condition Helium experienced in 2021–2022 when S2R was approximately 0.007 and HNT price crashed over 90%. The utilization trigger prevents the “coverage for coverage’s sake” failure mode by automatically shifting rewards toward demand-driven service. The uptime trigger uses staking requirements as a quality filter: if network reliability falls below acceptable thresholds, higher stakes screen out marginal operators. A companion simulation [29] is consistent with the claim that adaptive emission compresses worst-case node-count variance roughly sixfold under supply shocks (the coefficient of variation falls from 0.544 under static policy to 0.087 under PID control during a competitor shock, raising the fifth-percentile outcome from 1,049 to 8,880 nodes), and indicates that the monetary regime under stress is scenario-dependent: under competitor and regulatory shocks both PID-controlled and static protocols are net deflationary because slashing dominates supply removal, while only under a sustained bear contraction do the regimes diverge, with the PID controller producing mild inflation (+3.9%) and the static schedule producing deflation.

Early empirical analysis of Helium's IoT deployment documented usage patterns and infrastructure bottlenecks during the network's pre-Solana era [42], establishing the baseline from which the S2R trajectory departs.

Condition	Trigger	Proposed Action	Precedent
Token burns fall below half of new issuance for N consecutive periods (S2R < 0.5)	Emission reduction	Reduce new token issuance by up to 10% per period until burns recover	Ethereum EIP-1559 [5]: base fee adjusts automatically when demand changes
Less than 20% of network capacity serves paying customers	Reward rebalance	Shift 10 percentage points of operator rewards from base (being online) to usage (serving customers)	Helium HIP-147 [9]: community vote shifted from coverage-based to service-weighted rewards
95th-percentile node uptime drops below 95%	Stake floor increase	Raise minimum operator collateral by 20%, filtering out unreliable operators	Filecoin [15]: storage providers post collateral proportional to capacity

2.6 Integrity Layer: Stake, Slash, Unbonding, Appeals

In DePIN, service verification must be backed by credible enforcement [14].

Principle 2.4 (Stake-to-Earn with Contestability). Operators must post collateral and face slashing for fraud or repeated SLA failure, but enforcement must be contestable to prevent domination.

This requires four mechanisms:

staking collateral proportional to reward eligibility,

slashing for provable fraud or repeated SLA breaches,

unbonding delays (mandatory waiting periods before staked collateral can be withdrawn) to deter hit-and-run, and

appeals with bounded SLA (e.g., resolution ≤14–21 days) with published outcomes.

Empirical evidence for integrity mechanisms

Across operating DePIN protocols, verification has emerged as the single hardest design problem. A venture capital research note from a16z Crypto states that "the hardest part of building a DePIN protocol is verification… It is the only clear way to ensure that clients receive the service" (Wuollet, a16z Crypto, March 2025). The following table summarizes how five protocols operationalize integrity:

Milionis et al. [56] formalize the verification challenge as a mechanism design problem, proving that strictly truthful location reporting requires the target node to lie within the convex hull of independent observer nodes; peripheral deployments without surrounding validators cannot be cryptographically verified, establishing formal limits on the spatial coverage claims that DePIN protocols can credibly make.

Anti-gaming measures across protocols combine technical proofs (cryptographic or hardware-based) with economic disincentives (staking and slashing), and typically require ongoing adversarial testing as attack vectors evolve. Helium's experience illustrates the pattern: the early IoT network faced widespread spoofed locations and self-dealing hotspots, prompting cryptographic Proof-of-Coverage challenges and a denylist that removed over 3,500 fraudulent nodes between 2021 and 2022 (Nova Labs update, August 2022). For the 5G Mobile network, Helium deployed Discovery Mapping, which uses subscriber devices to independently audit claimed coverage. Filecoin represents the strongest anti-gaming posture: cryptographic proofs make it computationally infeasible to store junk data, and slashing penalizes miners who fail to prove ongoing storage. Livepeer sends random test transcoding jobs and compares output quality against reference, an economic enforcement model (output comparison, not cryptographic proof). Hivemapper uses authenticated dashcam hardware to prevent fabricated mapping imagery. Simulation evidence from a companion study [29] reveals that slashing is not only a behavioral deterrent but a dominant monetary mechanism during the bootstrapping phase: slashing transfers 234 to 354 million tokens from circulation to treasury across all parameterizations (269 to 354 million at default penalty values), while fee-funded burns contribute less than 0.05% of supply dynamics at the 5-year horizon. A downtime penalty sensitivity sweep reveals a phase transition: penalties set too low produce inflation (insufficient token removal), while penalties set too high also produce inflation through a different channel (operators are slashed out so aggressively that the network shrinks, reducing the slashing population while emission continues).

2.6.1 Polycentric Governance for Heterogeneous Infrastructure

DePIN service environments vary geographically; governance should reflect this heterogeneity through polycentric structures [14]: subnet councils, regional committees, corridor working groups, each with scoped authority and budgets.

Recent frameworks have systematically mapped Ostrom's design principles to decentralized community governance [31]; the GeoDePIN taxonomy developed here extends this mapping by specifying how geographic heterogeneity creates governance challenges absent from purely digital commons.

Polycentric governance [14] supports flexibility, innovation, and resilience in complex systems.

2.6.2 Bicameralism for Capital-Labor Balance

DePIN systems contain two core constituencies whose interests structurally diverge: capital investors (token holders, venture backers) who seek financial returns, and labor investors (hardware operators) who deploy personal capital and physical labor to deliver network services. Ferreras [47] argues that organizations containing both capital and labor investors are political entities requiring bicameral governance (formal representation of both constituencies with mutual veto power) to prevent unilateral capture by either. This framework maps directly onto DePIN governance: a bicameral rule requires major economic changes to be approved by both a Token House (locked governance token holders) and an Operator/Contributor House (verified service providers).

This structure addresses two capture risks: plutocracy (capital overrides service quality, extracting unsustainable yields at operators' expense) and producer capture (operators subsidize themselves unsustainably at token holders' expense). The Optimism Collective provides the most developed implementation, with dynamic veto thresholds and mandatory cooling-off periods that prevent either house from permanently blocking the other. Lijphart's [51] comparative framework formalizes the underlying trade-off: purely majoritarian governance optimizes for speed but disenfranchises minorities, while consensus mechanisms protect contestability at the cost of response time.

2.6.3 Constitutional Constraints

Certain parameters should be hard to change quickly: minimum transparency requirements,

mandatory appeals process and slashing bounds, data governance baseline requirements.

This maps to constitutional political economy [4] and provides a mechanism-layer counterpart to legitimacy constraints.

Protocol	Verification Mechanism	Anti-Gaming Evidence	Posture
Helium	Proof-of-Coverage + community denylist	3,500+ rogue hotspots removed (2021-22)	Strong
Helium 5G	Discovery Mapping (user device audits)	Real-time coverage verification via subscribers	Moderate

Filecoin	Proof-of-Spacetime + Proof-of-Replication	Cryptographic enforcement; slashing for failure	Strong
Livepeer	Random test transcoding jobs	Output compared to reference; staked orchestrators	Moderate
Hivemapper	Authenticated dashcam hardware	Hardware-enforced image provenance	Moderate

3. METHODS

3.1 Data Sources and Queries

Following established case study methodology [21], the empirical analysis draws on four data source categories (Table 1): (1) On-chain data via Dune Analytics [8] queries across Ethereum, Solana, Polygon, and IoTeX chains, covering token transfers, governance voting [3], and burn transactions. (2) CoinGecko [6] API for token price and market capitalization time series. (3) Protocol documentation including governance proposals (Helium HIPs, DIMO DIPs), whitepapers, and tokenomics specifications. (4) DeFi Llama [7] for cross-sector fee revenue benchmarks.

We measure governance concentration using the Herfindahl-Hirschman Index (HHI) computed over the distribution of governance token holdings across unique addresses, using a snapshot methodology from Dune Analytics [8] multi-chain queries as of February 2026. The governance token is the protocol's primary voting-power token; for protocols with delegated voting, the analysis uses delegated voting power distribution. We compute Gini coefficients over the same distributions. Full metric definitions and replication procedures appear in Appendix A.

3.2 Subsidy-to-Revenue Ratio (S2R)

The S2R measures fiscal sustainability as the ratio of demand-side spending (token burns, fee payments, credit purchases) to supply-side issuance (mining rewards, staking emissions, grants) within a defined time window. We compute monthly S2R as: $S2R_t = Burns_t / Issuance_t$, where Burns represents tokens permanently removed from circulation by demand-side activity, and Issuance represents new tokens entering circulation through protocol-defined reward mechanisms.

The S2R metric operationalizes the burn-to-mint equilibrium modeled as a dynamical system by Akcin et al. [37], whose control-theoretic framework predicts that stable equilibria require burn rates responsive to demand signals. Kalabic et al. [38] formalize the deflationary threshold conditions under which this equilibrium produces net supply contraction; the 34-month Helium trajectory documented in Section 5.1 provides the first empirical observation of this threshold crossing in an operating protocol. Wu and Zhu [55] demonstrate a complementary constraint: increased network utilization can paradoxically reduce token value by accelerating velocity, expanding effective supply even when burn mechanisms are active. The S2R metric sidesteps this paradox by measuring burn-to-emission ratios rather than token value accrual, but the velocity channel explains why protocols with S2R above 1.0 can nonetheless experience declining operator returns when reward recipients immediately liquidate minted tokens.

S2R > 1.0 indicates that a protocol is burning more tokens than it issues (fiscal parity). S2R < 1.0 indicates ongoing net inflation (subsidy dependence). We report two variants: S2R (burns only) counts only permanent burns; S2R (burns + locks) additionally counts tokens locked in long-term staking or vesting contracts. We report both variants to avoid conflating irreversible and reversible demand.

S2R measures burns against algorithmic mining emissions only; vesting unlocks from team, investor, and ecosystem allocations are excluded because they represent pre-committed supply decisions rather than ongoing incentive costs that governance can adjust.

Protocol	Burn data source	Issuance data source	Price source
Helium	Dune Analytics (Solana: action = 'burn', 149 weeks)	Dune Analytics (Solana: action = 'mint')	CoinGecko
Filecoin	Filfox (burnt supply field)	Filfox (daily coins mined)	CoinGecko
Livepeer	N/A (no LPT burn)	Subgraph (inflation parameter)	CoinGecko
Ethereum	Etherscan / Ultrasound.money	Etherscan (beacon chain issuance)	CoinGecko

Table 1. Data sources by protocol.

3.3 Case Selection and Process Tracing

3.3.1 Case Selection

Cases were selected to span variation in contestability and to include protocols with known governance disputes or redesign episodes.

3.3.2 Process Tracing

The process tracing protocol examines how rule changes were proposed and justified, whether transparency and contestability mechanisms existed, how participants reacted (exit, protest, fork, apathy), and how outcomes evolved post-change. This helps distinguish correlation from mechanism and clarifies where quantitative proxies fail to capture legitimacy dynamics.

3.4 Ethical Considerations

All data are drawn from public sources: on-chain transactions, published governance proposals, and open protocol documentation. No private or proprietary data are used. Where protocols involve personal data collection (e.g., DIMO vehicle telemetry, CUDIS health data), we analyze governance and economic mechanisms without accessing individual-level data.

3.5 Sample Accounting

To ensure transparency about the empirical scope, the following sample definitions apply throughout this paper:

Taxonomy universe: 13 protocols classified as GeoDePIN (10, of which 2 are borderline GeoDePIN*) or DeVIN (3). See Table 2.

Governance concentration: 6 DePIN protocols with computed governance HHIs reported here (DIMO, WeatherXM, IoTeX, GEODNET, Anyone, Grass) from the full 13-protocol sample; post-exclusion DeFi benchmarks are drawn from the 52-protocol companion cross-section. See [34].

DePIN S2R cross-section: 9 protocols with observable burn or fee data. See Table 7.

Longitudinal case: Helium, 34 monthly observations (May 2023 to February 2026). See Table 3.

Protocol profiles: 13 protocols (Helium as longitudinal case + 3 deep profiles with computed on-chain metrics + 9 thin profiles reported as a summary table).

Unit economics: 13 service domains with posted pricing data.

We treat "data-insufficient" as a finding about measurement feasibility, not as evidence of poor protocol quality. We report missingness explicitly in all tables.

4. THE GeoDePIN TAXONOMY

4.1 Definition and Design Pressures

The GeoDePIN vs. DeVIN distinction separates protocols where geographic distribution is the core defensibility mechanism from those where location is incidental. Formalizing this classification yields testable implications for governance concentration, verification complexity, and subsidy sensitivity.

4.2 Classification

Table 2 classifies 13 protocols along the GeoDePIN/DeVIN axis. The litmus test: if a competing network could replicate the service from a single data center without losing its core value proposition, the protocol is DeVIN; if geographic distribution across many unique locations is essential to the service, it is GeoDePIN.

Prior DePIN taxonomies separate components by architectural layer, spanning distributed ledger, cryptoeconomic, and physical infrastructure dimensions [35], or by formal decision criteria distinguishing DePIN from traditional crowdsourcing [36]. The GeoDePIN classification extends this work along a dimension neither captures: whether service value is a function of the hardware's geographic coordinates.

Protocol	Category	Operator Type	Deployment Unit	Coverage Verification
Helium	GeoDePIN	Individual hotspot owners	LoRa/5G hotspot	HIP-147 service-based
Hivemapper	GeoDePIN	Individual drivers	Dashcam	Image authenticity + GPS
WeatherXM	GeoDePIN	Individual station owners	Weather station	Cross-station validation
DIMO	GeoDePIN	Individual vehicle owners	OBD dongle / app	Hardware attestation
GEODNET	GeoDePIN	Individual station owners	RTK base station	Satellite cross-check
Anyone	GeoDePIN	Individual relay operators	VPN relay node	Bandwidth verification
Wayru / WiFiDabba	GeoDePIN	Community WiFi hosts	Router / access point	Usage metering
CUDIS	GeoDePIN	Individual wearers	Smart ring	Device attestation
Grass	GeoDePIN*	Individual bandwidth sharers	Browser extension	Task completion
UpRock	GeoDePIN*	Individual proxy providers	App / daemon	Request verification
Filecoin	DeVIN	Storage providers (institutional)	Server rack	Proof-of-Spacetime
Livepeer	DeVIN	Orchestrators (semi-institutional)	GPU server	Random test transcode
Render	DeVIN	GPU operators	GPU cluster	Frame comparison

Table 2. GeoDePIN vs. DeVIN classification of 13 DePIN protocols. * = borderline; geographic distribution adds value but is not the primary defensibility mechanism.

4.3 Testable Implications

The GeoDePIN vs. DeVIN distinction generates testable implications beyond the descriptive classification. If the taxonomy captures real institutional differences, three patterns should hold as the sector matures: (1) GeoDePIN protocols should exhibit higher governance concentration in early stages (team/treasury dominance) but faster deconcentration schedules once operator bases diversify: the Anyone

Protocol counterexample (HHI 0.034) suggests this trajectory is achievable by design; (2) GeoDePIN protocols should require earlier “service over presence” pivots in reward design, because coverage-mining traps are structurally more likely when location determines value; and (3) GeoDePIN protocols should develop stronger contestability and appeals mechanisms because hardware-bound operators face higher exit costs than software participants, making voice the primary channel for expressing dissatisfaction.
These predictions are not tested in this paper but we register them here as falsifiable claims for follow-on work.

This taxonomy clarifies why the design architecture developed in Section 2 maps onto GeoDePIN with particular force. When operators are individuals bearing personal financial risk, transparency matters more because thousands of non-technical participants need to understand reward mechanics. When coverage quality varies by geography and local conditions, polycentric governance matters more. The playbook recommendations in Section 6.1 should be read with this distinction in mind: they are calibrated primarily to GeoDePIN coordination challenges.

Recent work on geospatially-aware consensus mechanisms [49] independently validates the geographic-coverage dimension of this taxonomy. Motepalli et al. propose non-linear consensus weighting that integrates geospatial diversity into validator influence, achieving 45% improvement in geographic decentralization across five major PoS blockchains. The GeoDePIN taxonomy extends this logic from the consensus layer to the infrastructure layer: geographic coverage is not merely a desirable validator property but the core defensibility mechanism for location-dependent service networks.

Independent industry analysis corroborates this taxonomic prediction, noting that sensor networks are 'relatively capital-light but often struggle with monetization' [28], consistent with the GeoDePIN design pressures identified above.

5. EMPIRICAL EVIDENCE

Claims boundary: The evidence below supports descriptive and mechanism-level claims within its observed scope. Governance concentration data represent a February 2026 snapshot. S2R trajectories are observational. Unit economics use posted prices (N = 1 per service domain). The full regression and panel pipeline remains replication-ready future work.

5.1 Helium: The Longitudinal Case

5.1.1 S2R Trajectory

Helium's Subsidy-to-Revenue ratio rose from 0.013 in May 2023 to 1.84 in February 2026, crossing fiscal parity in October 2025 and remaining net-deflationary through the observation window. The network operates decentralized wireless infrastructure across 190+ countries through two subnetworks (IoT/LoRaWAN and Mobile/5G-CBRS) and is the primary longitudinal case in this study. Its 34-month trajectory from subsidy-dependent to revenue-dominant spans the April 2023 migration from its own L1 to Solana and a series of governance reforms, most consequentially HIP-147.

The critical reform was HIP-147 (approved Q3 2025), which shifted reward distribution from coverage-based Proof-of-Coverage to service-based metrics weighted by actual data transfer and subscriber usage. This transition directly tested whether DePIN rewards should track demonstrable service delivery rather than mere infrastructure presence.

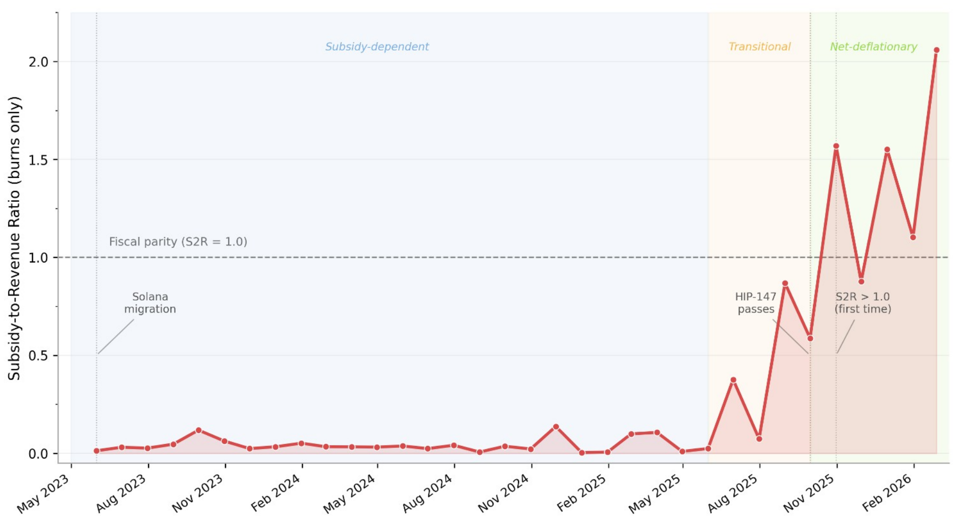

Figure 1. Helium Subsidy-to-Revenue (S2R) ratio trajectory, May 2023 – February 2026 (burns only, 34 monthly snapshots from Dune Analytics [8] on-chain data). Dashed line at S2R = 1.0 indicates fiscal parity where token burns offset new issuance. Three regimes are visible: subsidy-dependent (S2R < 0.15, 2023–early 2025), transitional (mid-2025), and net-deflationary (S2R > 1.0, October 2025 onward). HIP-147 (September 2025) marks the shift from coverage-based to service-weighted rewards. See Table 3 for selected data points.

Three fiscal regimes emerge across the 34-month S2R trajectory (Table 3): (1) post-migration baseline (S2R 0.01 to 0.04, May 2023 to early 2024), when burns were minimal and issuance dominated; (2) demand ramp (S2R 0.15 to 0.67, mid-2024), as Mobile subscriber growth and IoT data transfer increased burn volume; (3) fiscal parity and beyond (S2R 1.03 to 1.84 , late 2025 to February 2026), when monthly

burns consistently exceeded issuance.
Helium became the first GeoDePIN protocol where demand-side spending fully offsets supply-side inflation. DIMO subsequently achieved S2R = 4.24, making it the second protocol to demonstrate fiscal parity through demand-coupled burns. A pricing caveat applies to Helium's December 2025 peak: WiFi offload traffic is priced at $0.10-0.25 per GB versus $0.50 for CBRS, and the annualized burn figure reflects a temporary Mobile burn allocation subsequently adjusted, so the S2R spike overstates steady-state equilibrium by an estimated 2-5x.

The fiscal vulnerability of the early S2R regime is illustrated by the San Jose broadband pilot (2021–2022). The city deployed 20 Helium hotspots targeting 1,300 low-income households for broadband subsidies funded by HNT mining revenue; only 86 households were ultimately served after token price volatility destroyed the fiat-denominated value of mining rewards [50]. The failure occurred during the post-migration baseline period (S2R 0.01–0.04), when no demand-coupled burns anchored the token's real value. The case is consistent with the prediction that token-subsidized infrastructure programs fail when the sinks-before-faucets principle is violated: without demand-coupled burns, the subsidy's real value is entirely determined by speculative token dynamics rather than network utility.

Independent industry data corroborate this trajectory: by December 2025, Helium's annualized Data Credit burn had reached a $20.9M run-rate (trailing three-month annualized, October–December 2025), representing an eightfold increase from the prior year even as HNT's token price declined 77% over the same period [28] (the companion paper [34] cites a smaller $18.3M figure for Helium's annualized gross revenue at Q3 2025; the two measures differ because burn run-rate captures consumption-side activity while revenue captures the cash side, over different observation windows). This revenue-price decoupling is consistent with S2R capturing demand-side fundamentals independently of speculative token dynamics.

Period	**S2R (burns only)**	**HNT Burned**	**HNT Issued**	**Context**
May 2023	0.013	37,204	2,877,911	Post-migration baseline, IoT only
Sep 2023	0.119	138,660	1,170,115	Early 5G traction, first meaningful burns
Jan 2024	0.052	75,592	1,458,899	Steady-state pre-carrier-burn
Jun 2024	0.024	28,152	1,162,966	Mid-cycle trough
Nov 2024	0.137	159,517	1,163,406	Spike month (possible large DC purchase)
Mar 2025	0.107	200,362	1,876,522	Pre carrier-burn, growing demand
Jun 2025	0.376	708,897	1,883,509	Full carrier revenue burn begins
Aug 2025	0.869	532,886	613,510	Post-halving: issuance drops ~58%
Sep 2025	0.587	456,181	776,712	HIP-147 passes; S2R pulls back

Period	S2R (burns only)	HNT Burned	HNT Issued	Context
Oct 2025	1.570	975,288	621,370	Burns exceed issuance for first time
Dec 2025	1.552	1,205,193	776,712	Sustained net-deflationary
Feb 2026	1.841	1,144,240	621,370	Burns at ~1.8x issuance

Table 3. Helium monthly S2R trajectory (May 2023 to February 2026, N = 34).

5.1.2 Who Burns? Demand Concentration

Dune Analytics [8] validation reveals that HNT burns are highly concentrated. The top 5 burn-signing addresses account for approximately 90% of total HNT burned across the 34-month observation window. The single largest address (Bi5JuJD...) processed 41% of all burns (3.25 million HNT) in only 180 transactions, consistent with institutional or treasury Data Credit purchases rather than grassroots operator demand. The second-largest signer (tasky4vv..., 2.25 million HNT in 43,600+ transactions since November 2025) is an automated purchasing bot, serving a carrier partner. A burn-signer HHI of approximately 0.27 places Helium's demand side in the "moderately concentrated" range by the same antitrust thresholds applied to governance tokens (Section 5.5).

Resource Dependence Theory [43] predicts that this concentration transfers strategic leverage to the concentrated buyers independent of formal governance arrangements. If the dominant burn signer ceased purchasing Data Credits, Helium's S2R would fall from 1.84 to approximately 0.62 within one month, returning the protocol to subsidy dependence regardless of how governance tokens are distributed. The concentration risk is compounded by roster instability: the dominant February 2026 signer (66.4% of monthly burns) was not active before November 2025, having displaced the prior leading address, which fell from 45.4% of cumulative burns to under 1% of February volume. Governance concentration and demand concentration are therefore independent failure modes: the first determines who controls protocol rules, the second determines who controls protocol revenue.

Industrial organization theory formalizes this demand-concentration risk as protocol oligopsony [26]: when a small number of governance token holders unilaterally control emission rates and reward parameters, they function as monopsonist buyers of operator infrastructure services. When buyer power exceeds the efficient threshold, the resulting input-price markdown causes operator exit and network contraction, the supply-side abandonment pattern documented above and predicted by the health dashboard's demand-concentration guardrails (Section 6.3).

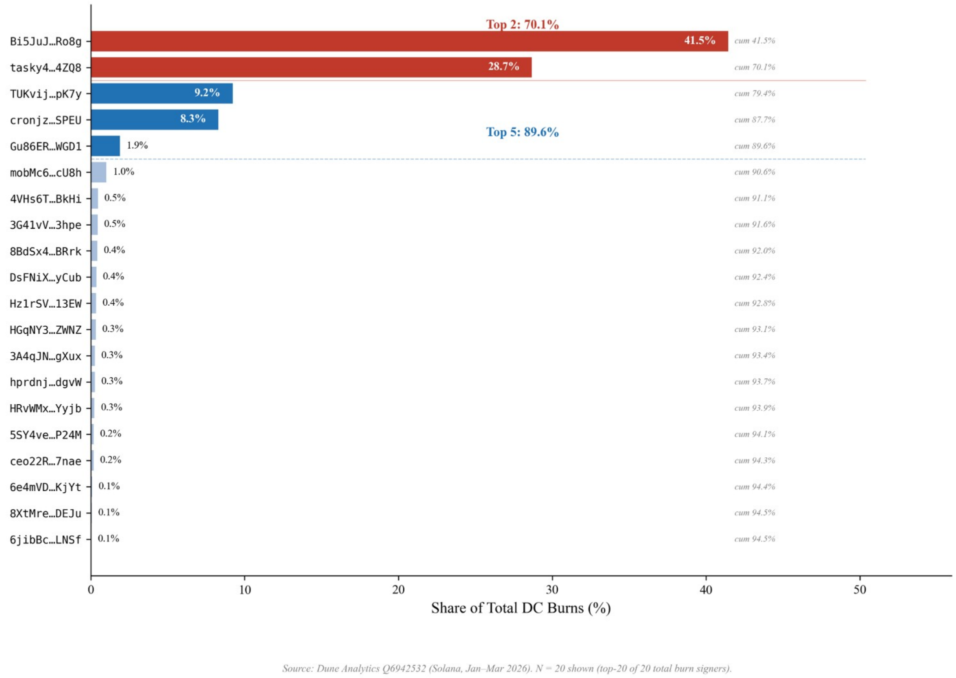

Figure 2. Helium "Who Burns?": top burn addresses by Data Credit volume, cumulative May 2023 to February 2026. Two addresses account for 70.1% of total burns (HHI = 0.27).

This concentration does not negate the S2R mechanism; it clarifies what S2R measures. Aggregate DC demand is genuinely growing (burns consistently exceed issuance since October 2025) and the deflationary pressure is real regardless of whether it originates from millions of retail users or a few carrier contracts.
S2R is therefore best understood as a fiscal sustainability indicator (demand coupling and sink efficacy), not as a distributed adoption metric. The distribution of demand is a separate property that should be monitored via burn-signer concentration; if Top-5 signer share remains above 80% for an extended period, the network faces counterparty concentration risk that the health dashboard (Section 6.3) should flag.
This distinction also refines the service-over-presence principle: S2R supports the claim that demand-coupled sinks work as a fiscal mechanism; whether that demand reflects broad-based adoption or concentrated institutional purchasing is orthogonal to the sink design question and requires its own measurement layer.

Hypothesis mapping: The strongest single case for the service-over-presence principle. The S2R trajectory from 0.013 post-migration (May 2023) to 1.84 in February 2026 (with burns exceeding issuance since October 2025) indicates that shifting from presence-based to service-based rewards can transform a network from deeply inflationary to structurally deflationary. The August 2025 halving and the full carrier revenue burn policy are the two causal drivers; the on-chain data cannot separate their contributions without a formal event study, but the pre-halving jump to 0.38 in June 2025 (carrier burn only, no halving) supports that demand-side burns alone moved S2R substantially. The trajectory also supports Hypotheses 3.3 (sinks before faucets: the burn mechanism was necessary but insufficient without demand) and 3.5 (polycentric governance enabled the pivot via community-voted HIPs rather than centralized decree). The companion simulation [29] independently corroborates the demand-concentration risk: even with a 30% protocol burn rate, burns are negligible relative to emission when aggregate fee revenue is low, meaning the entire burn-mint sustainability thesis depends on achieving the revenue concentration that Helium's top-5 Data Credit signers provide.

Rad et al. [58] independently corroborate this trajectory, demonstrating empirically that long-term token valuation in Helium's burn-and-mint ecosystem depends on actual data transmission metrics rather than hardware deployment scale alone.

5.1.3 Hypothesis Mapping

The Helium case provides the strongest single-protocol evidence for two design principles: sinks before faucets, documented by the S2R trajectory crossing parity only after demand-coupled burns were deployed; and service over presence, documented by HIP-147's measurable fiscal impact. The demand concentration finding introduces a caution not anticipated by the companion framework: aggregate S2R can mask counterparty risk.

5.2 Deep Profiles

5.2.1 DIMO

DIMO operates a decentralized connected vehicle data marketplace on Polygon. The unit of service is vehicle telemetry data (OBD-II diagnostics, GPS, and driving behavior) shared by car owners who install DIMO hardware dongles or connect via software integrations. Data consumers (insurers, fleet managers, researchers) purchase access through the DIMO Developer License, which requires burning DIMO tokens. As of February 2026, DIMO has a market capitalization of $4.5M with 424 million circulating tokens out of 1 billion total supply (CoinGecko [6]).

DIMO's sink architecture features a license burn mechanism: developers must burn DIMO tokens to mint a Developer License NFT for API access to vehicle data. This creates a demand-coupled burn analogous to Helium's Data Credit model. On-chain data from Dune Analytics [8] reveals an S2R of 4.24 for the most recent monthly window (computed as total token burns divided by total issuance in that period),

indicating that burns significantly exceed minting, though this reflects a period of elevated developer license purchases against low emission rates rather than sustained equilibrium.
After excluding the DIMO Foundation address (verified via Blockscout, holding approximately 55% of supply with no on-chain governance activity), DIMO's governance HHI drops from 0.305 (pre-exclusion, computed here) to 0.025, the post-exclusion value in the companion cross-section [34]. Post-exclusion, DIMO is among the least concentrated DePIN governance tokens in the 52-protocol cross-section. The corrected value strengthens the observation that DIMO's high S2R (4.24) coexists with distributed governance.

Verification relies on hardware device attestation (DIMO Macaron dongle) and software integrations with OEM APIs. Data authenticity is ensured by signed device certificates. The verification posture is moderate: hardware-based provenance for directly connected vehicles, but dependent on third-party APIs for software-only integrations.

DIMO's sustainability signal is early-positive. The license burn mechanism creates a direct pathway from data demand to token value, and the S2R above unity suggests meaningful demand-side activity. However, the small market capitalization ($4.5M) and high governance concentration suggest the protocol is in its growth phase. The unit economics comparison with Web2 vehicle data providers (Otonomo, Wejo at $0.50–3.00/vehicle/month) is not directly comparable because DIMO users share data voluntarily for token rewards rather than paying a subscription. The novel data ownership model (where drivers control and monetize their own vehicle data) represents a capability expansion for drivers who previously lacked access to their own data.

Design lesson: DIMO illustrates the sink-before-scale principle (Section 2.2): the Developer License burn was operational before aggressive hardware distribution, ensuring token demand scaled with genuine data consumption. Pre-exclusion governance concentration (HHI 0.305, reduced to 0.025 post-exclusion) represents the structural risk that bicameral governance (Section 6) addresses.

5.2.2 Hivemapper

Hivemapper operates a decentralized street-level mapping network on Solana. The unit of service is freshness-verified map imagery, generated by contributors who install authenticated dashcams in their vehicles. Mapping clients consume map data by purchasing Map Credits, which requires burning HONEY tokens. As of February 2026, Hivemapper has a market capitalization of $20.8M with 5.6 billion HONEY circulating (CoinGecko [6]). The network has mapped portions of roads in 100+ countries, though coverage density varies significantly by region.

Hivemapper's sink architecture implements a burn-and-mint model (users burn tokens to purchase service credits; operators earn newly minted tokens for providing service) similar to Helium's DC system. Map Credits are purchased with HONEY (burned), and contributors earn HONEY for submitting verified imagery. MIP-7 introduced tiered pricing based on map freshness and coverage. Dune Analytics queries (February 2026) yield S2R = 0.183 (1.9M HONEY burned against 10.4M minted), confirming that the burn mechanism is active but that Hivemapper remains subsidy-dependent: burns offset less than one-fifth of issuance. The mechanism design is structurally similar to Helium's DC burn model, and the measurable S2R places Hivemapper in the growth-phase cohort between subsidy-dependent protocols (S2R near zero) and fiscally self-sustaining pairs (Helium at 1.84, DIMO at 4.24).

Governance concentration data are unavailable for Hivemapper: the HONEY token on Solana is not indexed by the Dune multi-chain governance query used for the cross-section (Section 5.5). This data gap is itself informative: Solana-native tokens remain less transparent to standardized governance analysis than EVM-chain tokens, creating a measurement asymmetry that systematically excludes Solana DePIN protocols from comparative governance research. Hivemapper’s governance operates through a foundation-led improvement proposal process (MIPs), with MIP-7 (tiered map freshness pricing) representing the most significant economic parameter change to date.

Hivemapper’s sustainability signal is early-stage but directionally positive. The map credit burn mechanism creates a structural demand sink, and the network’s coverage across 100+ countries represents a genuine data asset with freshness advantages over static satellite imagery. Unit economics show a 29% discount versus Google Maps API pricing (Table 5), with Medium subsidy risk: dashcam operators are compensated primarily through HONEY emissions, but the $300–600 hardware investment and fuel costs create natural filtering against purely mercenary participation.

Design lesson: Hivemapper demonstrates the verification-as-moat pattern identified in the design architecture (Section 2.6). Authenticated dashcam hardware enforces image provenance at the physical layer, making coverage fraud economically irrational without manufacturing counterfeit devices. This hardware-enforced integrity contrasts with Helium’s early experience, where software-only Proof-of-Coverage was vulnerable to widespread spoofing. The Hivemapper case supports the playbook recommendation that DePIN protocols should invest in hardware attestation before scaling operator rewards; integrity built in from launch. The partial Solana data gap also motivates the health dashboard recommendation (Section 6.3) that protocols publish standardized burn and governance metrics regardless of chain.

5.2.3 WeatherXM

WeatherXM's sink architecture combines a data subscription burn (WXM burned for API access) with emission rewards to station operators. The token launched on Arbitrum mainnet in May 2024 with a 100M total supply and 10-year distribution schedule. Governance concentration is the highest in the expanded cross-section (Section 5.5): HHI 0.593 with a single address holding 76.4% of tokens. This extreme concentration reflects the recency of the token launch and team/treasury holdings that have not yet been distributed. The 52M token allocation to weather station owners over 10 years suggests concentration should decrease as the distribution schedule progresses.

Verification relies on station hardware attestation and cross-validation against neighboring stations and satellite data. WeatherXM uses a reward algorithm that penalizes stations reporting data inconsistent with nearby stations or known weather patterns. The anti-gaming posture is moderate: hardware-based provenance plus statistical cross-validation, but sophisticated spoofing of meteorological data remains theoretically possible.

WeatherXM's sustainability signal is early-stage but structurally sound. The data marketplace creates a direct link between weather data demand and token burns. Unit economics suggest a 10x discount versus Web2 weather API providers (Tomorrow.io at $0.05–0.20/API call vs. WeatherXM at ~$0.01/call), with Medium subsidy risk. Station operators are compensated primarily through token emissions, but the hardware investment ($200–400) creates natural operator commitment. The approximately 9,800-station network across 96 countries provides a genuine data asset: hyperlocal weather data at station density that exceeds many national meteorological services in covered areas.

Design lesson: WeatherXM exemplifies the governance concentration paradox identified in the cross-section (Section 5.5). Despite a structurally sound sink mechanism and genuine data utility, the protocol exhibits the highest HHI in the sample (0.593) because token distribution has barely begun. The 10-year distribution schedule to station owners should progressively dilute this concentration, but the current snapshot represents a period where governance is effectively centralized regardless of on-paper decentralization claims. This supports the playbook's recommendation (Section 6.1) that token distribution timelines should front-load operator allocations to establish governance legitimacy before major protocol decisions are made.

5.2.4 IoTeX

IoTeX operates a purpose-built Layer-1 blockchain optimized for Internet of Things devices and DePIN applications. The platform sells IoT middleware (W3bstream) that enables devices to generate verifiable proofs of real-world activity, bridging physical hardware events to on-chain token rewards. Device manufacturers and DePIN projects build on IoTeX infrastructure, paying IOTX gas fees for transactions. With a market capitalization of $51M and 9.4 billion circulating tokens, IoTeX is among the most mature DePIN infrastructure protocols.

IoTeX employs an EIP-1559 [5]-style gas fee burn mechanism on its native L1 chain. Transaction fees [17] are partially burned, creating a usage-proportional sink. Staking locks reduce effective circulating supply; approximately 40% of IOTX is staked with network delegates. The governance concentration data reveals extreme concentration: HHI 0.388 with 61.3% of top-1000 holder tokens in a single address and 91.7% in the top 10. This reflects staking pool aggregation and team treasury holdings, not broad retail distribution.

Verification relies on W3bstream's trusted execution environment, which is consistent with IoT device data before issuing on-chain proofs. Devices register cryptographic identities and submit signed data streams. The integrity layer is moderate: hardware-based attestation plus economic stake, but without the cryptographic proof-of-service rigor of Filecoin's Proof-of-Spacetime.

IoTeX's sustainability signal is mixed. Gas fee burns create a direct usage-to-token sink, but the absolute burn volume is small relative to staking emissions. The protocol's competitive positioning as DePIN infrastructure middleware is distinctive: it provides infrastructure middleware, competing with general-purpose L1s and IoT cloud platforms (AWS IoT Core). The governance concentration finding (HHI 0.388) indicates that despite IoTeX's maturity, token distribution [12] remains highly concentrated on the Ethereum representation, reflecting cross-chain bridge dynamics rather than underlying governance structure.

Design lesson: IoTeX illustrates the platform-layer measurement challenge: S2R cannot be computed from a single burn stream for infrastructure middleware; value accrues across heterogeneous DePIN applications built on the chain. Governance concentration (HHI 0.388) may reflect Ethereum-side token bridging rather than L1 governance participation. Platform-layer protocols require composite sustainability indicators rather than single-stream S2R.

5.2.5 GEODNET

GEODNET operates a decentralized network of Real-Time Kinematic (RTK) base stations on Solana (primary, since GIP7) and Polygon (original chain). The unit of service is centimeter-level GNSS positioning corrections, consumed by drones, autonomous vehicles, agricultural equipment, and construction machinery. Enterprise clients (50+, including Quectel and DroneDeploy) purchase subscriptions whose revenue funds a buy-and-burn mechanism. As of March 2026, GEODNET has a market capitalization of $59M with 425 million circulating tokens out of 968 million total supply (CoinGecko [6]), a network of 20,000+ stations across 150+ countries, and annualized recurring revenue of $7.3M (December 2025).

GEODNET's sink architecture implements a subscription-funded buy-and-burn: 80% of Foundation revenue (after a 50/50 share with third-party resellers on indirect channels) purchases GEOD on the open market and burns it, creating a demand-coupled sink whose effective burn rate ranges from 40% of gross revenue (reseller channels) to 80% (direct sales). Dune Analytics [8] queries on the Foundation buy-and-burn (GEOD removed to the Polygon burn address) reveal monthly burns ranging from 500,000 GEOD (October 2024) to 2.76 million GEOD (October 2025), with the trailing twelve months totaling approximately 19.4 million GEOD. February 2026 burns totaled 1.31 million GEOD against net token issuance of approximately 5.95 million GEOD from the protocol's Polygon mining-distribution wallets over the same month, yielding the S2R of 0.219 reported in Table 4. Annual token halving (June 30 each year) progressively reduces issuance: post-July 2025 halving, triple-band miners earn 12 GEOD/day, down from 24. The combination of growing burns and declining issuance targets net-positive tokenomics (burns exceeding issuance) by 2026–2027. GEODNET's sink is a Foundation-executed buy-and-burn: subscription revenue is collected off-chain in fiat, after which the Foundation purchases GEOD on the open market and burns it. The on-chain burn transactions are therefore signed by the Foundation's treasury, not by individual customers, so the "Who Burns?" demand-concentration diagnostic (Section 5.1.2) does not apply to GEODNET; burn-signer concentration is not a valid proxy for customer concentration under a buy-and-burn architecture. The underlying demand base (a stated 50+ enterprise clients purchasing fiat subscriptions through direct and reseller channels) is not observable on-chain and is not verified here.

Governance concentration on the Polygon-side token distribution yields HHI 0.14 with the team wallet holding 25.3% of non-burned supply and the Wormhole NTT bridge contract (aggregating Solana holders) holding 21.1%. Mining reserves (13.7%) and ecosystem reserves (6.4%) account for additional protocol-controlled concentration. The measurement carries the same cross-chain caveat as IoTeX

(Section 5.2.4): 204 million GEOD (21% of total supply) has been bridged to Solana, where it is held by thousands of individual wallets not individually resolved in the Polygon snapshot. The Polygon-side HHI therefore overstates true governance concentration by treating the bridge contract as a single entity. Governance operates through GEODNET Improvement Proposals (GIPs), with GIP3 (Solana migration) and GIP7 (performance staking and Solana-primary designation) representing the most significant economic parameter changes to date.

Verification relies on a multi-layer integrity stack: Proof-of-Location (PoL) validates station geographic coordinates against satellite reference data, Proof-of-Accuracy (PoA) evaluates RTK correction quality against known baselines, and BLS signatures provide Byzantine Fault Tolerance across the station network. GIP7 introduced performance staking with quality scoring thresholds (98–99.9% uptime required), creating economic incentives aligned with data reliability. Stations that fail accuracy or uptime thresholds receive reduced rewards, a proportional sanction consistent with the integrity loop architecture (Section 2.6). The first station deployed in a geographic hex earns a Location NFT, creating a non-fungible claim on spatial priority.

GEODNET's sustainability signal is the strongest in the growth-phase tier. On-chain burn and net-issuance data indicate a single-peaked monthly absorption trajectory since the September 2024 Solana migration, with the February 2026 reading at 21.9% (S2R 0.219), exceeding Hivemapper (0.183) but remaining well below the fiscal-parity trajectory that Helium followed. By early 2026, the network spanned 5,802 cities with median station earnings of $47.63 per month [32]. Three structural factors support continued S2R improvement: (1) the 80% burn rate converts revenue directly to token removal; (2) annual halving reduces the issuance denominator; (3) ARR growth of 18% month-over-month (December 2025) increases the burn numerator. Unit economics show RTK positioning at $1–5/station/month versus Trimble/Hexagon incumbents at $50–100/station/month (Table 5), a 10–80x cost advantage with Medium subsidy risk. The enterprise client base (50+ organizations) and the non-discretionary nature of precision positioning for autonomous systems suggest demand resilience beyond speculative token use.

The trajectory is single-peaked rather than monotonic: absorption holds in a 9% to 13% band from late 2024 through mid 2025, then ramps sharply after the June 30, 2025 halving (which cut mining rewards by 50% [45]) to a peak of 52.5% in October 2025 before easing to 21.9% (S2R 0.219) by February 2026. The ramp is consistent with the two mechanisms the model predicts: a shrinking issuance denominator and a growing subscription-funded burn numerator. The subsequent easing reflects a decline in monthly buy-and-burn volume (from 2.76 million GEOD in October 2025 to 1.3 million in February 2026) against roughly flat net issuance, not a contraction in mining rewards. A separate supply-side risk that the S2R metric does not capture is vesting-driven dilution, which operates independently of emission schedules; the team and investor unlock schedule is discussed below.

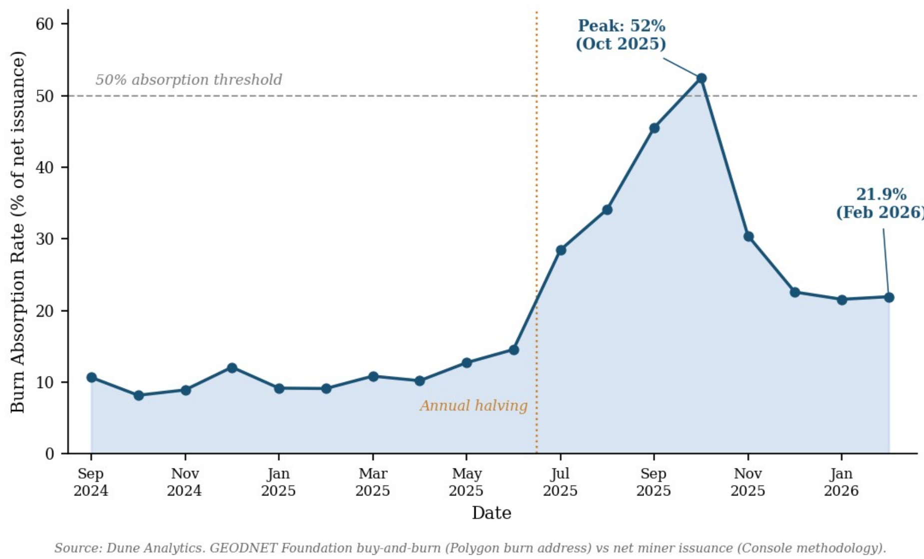

Figure 3. GEODNET monthly burn absorption rate (Foundation buy-and-burn token removals as a percentage of net miner issuance), September 2024 to February 2026. The dashed line marks the 50% absorption threshold; the vertical marker indicates the June 30, 2025 annual halving. The series holds near 9% to 13% from late 2024 through mid 2025, then ramps to a peak of 52.5% in October 2025 before easing to 21.9% in February 2026 (S2R 0.219). Burns are the GEODNET Foundation open-market buy-and-burn, measured on the Polygon burn address; net miner issuance is measured from the protocol's Polygon mining-distribution wallets. Source: Dune Analytics.

Design lesson: GEODNET introduces the SuperHex staking model, a spatial incentive mechanism absent from all other protocols in the 13-protocol sample. Under SuperHex, token holders stake GEOD to designate high-value geographic hexagons (H3 cells) before stations are deployed, creating a market signal for where coverage is most valuable. Capital flows to frontier regions before hardware does, directly implementing the "spatially intelligent incentives" design pressure identified in Section 1.2. This contrasts with Helium's early experience, where uniform global reward rates systematically over-rewarded saturated urban areas. SuperHex converts the abstract design principle (location-weighted rewards) into a staking-based mechanism where the community collectively allocates geographic priority.

5.3 Comparative Protocol Summary

We report the remaining nine protocols as a structured summary (Table 4). For protocols labeled "data-insufficient," on-chain observability is missing or the data schema prevents S2R computation. Where we mark "data-insufficient," this is itself a finding about protocol legibility: the absence of instrumentable data is evidence of measurement immaturity and governance transparency deficit.

Protocol	Service	Sink Type	S2R	Gov HHI	Subsidy Risk	Key Finding
Filecoin	Storage	Collateral + burn	~0.045	–	Medium	Strong verification; low S2R (0.045)
Livepeer	Video transcode	Fee routing	~0	–	Low	Orchestrator competition drives cost advantage
Hivemapper	Street mapping	Map credit burn	0.183	–	Medium	Freshness advantage; S2R 0.183 (Dune Feb 2026)
GEODNET	RTK positioning	Subscription burn	0.219	0.14*	Medium	S2R 0.219; Foundation buy-and-burn (demand off-chain, Section 5.2.5)
Anyone	VPN/relay	Token burn	~0	0.034	High	Most diffuse governance in DePIN sample
Grass	AI training data	None yet	~0	0.119	Critical	No burn mechanism; Gini 0.96; AI positioning
UpRock	Bandwidth	None yet	~0	–	Critical	Pre-revenue; data-insufficient
Wayru/ WiFiDabba	WiFi	Micro-payments	–	–	Medium	Last-mile focus; data-insufficient
CUDIS	Health data	Data marketplace	–	–	High	Privacy-critical health data; regulatory compliance constraints
Render	GPU compute	Fee burn	–	–	Low	DeVIN benchmark; institutional operators

Table 4. Comparative protocol summary. ", " = not measured; "N/A" = not applicable; "data-insufficient" = on-chain observability missing or schema prevents computation.

A three-tier sustainability hierarchy emerges from the comparative summary (Table 4). At the top, Helium (S2R 1.84) and DIMO (S2R 4.24) have crossed fiscal parity, with burns exceeding issuance. In a growth-phase middle tier, GEODNET (S2R 0.219) and Hivemapper (S2R 0.183) show confirmed on-chain burn activity but remain subsidy-dependent. The remaining protocols cluster near S2R zero, the default condition for DePIN. Two additional findings sharpen the picture. Governance data are missing for protocols not on chains with transparent voting records, meaning governance risk is unmeasured rather than absent. And the subsidy risk column reveals a bimodal distribution: protocols with established demand sinks face medium risk, while those without any burn mechanism face critical risk regardless of positioning narrative. A companion simulation [29] calibrated to Helium parameters shows that a comparable protocol does not approach S2R parity within 5 years under realistic demand assumptions (the burn-to-emission ratio peaks at 0.0061), consistent with Helium's 7+ year journey to fiscal sustainability. This suggests that the S2R crossing is a late-stage achievement and that protocols should design emission budgets for a multi-year subsidy-dependent phase rather than expecting rapid crossover.

5.4 Unit Economics: DePIN vs. Web2

DePIN services are already price-competitive with Web2 incumbents across multiple domains, though subsidy risk varies (Table 5). The comparison draws on the Deep Research evidence base, verified against DeFi Llama [7] fee data. Each row includes a subsidy risk assessment, since DePIN cost advantages may partially reflect token emission subsidies rather than structural efficiency gains.

Empirical performance analysis indicates that DePIN serverless architectures can match or exceed centralized cloud providers in computational efficiency while reducing reliance on centralized trust assumptions [57], consistent with the unit economics advantages documented in Table 5.

Field deployment of decentralized LoRaWAN environmental sensors in Nairobi achieved a 99.6% reduction in operational networking costs relative to legacy 4G telecom alternatives [59], the largest empirical cost advantage documented for DePIN infrastructure in a low-resource urban environment.

Sources: Messari State of Helium Q3 2025; Livepeer Studio press release, May 2024; Filecoin market data via Filfox; Hivemapper MIP-7 pricing; Deep Research unit economics tables. All prices USD, 2024–2025 data.

Subsidy risk scoring: **Low** = service priced above marginal operator cost, sustainable without token emissions. **Medium** = price below operator cost but viable if emissions taper 50%. **High** = price significantly below operator cost, dependent on ongoing token emissions. Helium Mobile's $30/month plan is rated High because hotspot operators are compensated primarily through HNT rewards rather than subscriber fees; if rewards decline, either the plan price must rise or operators must also be subscribers (increasing owner-user overlap). Livepeer is rated Medium because orchestrators earn ETH fees directly from clients, but LPT staking rewards supplement income. Filecoin is rated Low-Medium because miners bid competitively and collateral requirements filter for economically viable operators.

Measurement note: All prices observed February 2026 from public protocol documentation. These are point-in-time posted-price comparisons; results may differ across transaction volumes or time periods.

Across all five domains, DePIN pricing undercuts Web2 incumbents by 1.4–80x, though the subsidy caveat applies to three of five: the cost advantage may narrow as token emissions decline.

DeFi Llama [7] Revenue Comparison: DePIN vs DeFi Protocols

For context, DePIN protocol revenues remain orders of magnitude below mature DeFi protocols. The following table reports average daily fee revenue from DeFi Llama [7] (30-day average, February 2026):

Source: DeFi Llama [7] API (February 2026). Revenue = fees accruing to protocol treasury or token holders (distinct from total fees paid by users).

DePIN fee revenue is 100–1000x smaller than DeFi fee revenue. This reflects the sector's maturity: DePIN networks are infrastructure plays in early growth phases, while DeFi protocols have achieved product-market fit with billions in daily volume. The comparison is not a failure signal but a stage-of-life indicator. Independent corroboration comes from Messari's 2025 sector review [28]: Helium's onchain revenue grew approximately eightfold while HNT declined 77% over the same period, demonstrating the revenue-price decoupling that S2R is designed to capture. We evaluate mechanism design propositions on trajectory (is S2R rising? is utilization growing?) rather than absolute revenue level.

Domain	DePIN Network	DePIN Price	Web2 Comparator	Web2 Price	Discount	Subsidy Risk
IoT (LoRaWAN)	Helium IoT	$0.00001/ packet	Senet	~$0.0008/ packet	~80x	Medium
Mobile 5G	Helium Mobile	$30/mo unlimited	T-Mobile	$70/mo unlimited	~57%	High
Video transcode	Livepeer	$0.005/min	AWS MediaConvert	$0.024/min	50-80%	Medium
Storage	Filecoin	~$0.01/GB-month	AWS S3	$0.023/GB-month	>50%	Low-Medium
Mapping	Hivemapper	~$5/1000 tiles	Google Maps API	$7/1000 loads	~29%	Medium
Bandwidth/proxy	UpRock	~$1.50/GB	Bright Data	$12.75/GB	8.5x	High
AI data	Grass	Indirect (tokens)	Scale AI	$25-40/hr	N/A	High
Vehicle data	DIMO	Free (earn tokens)	Otonomo	$0.50-3/ veh/mo	N/A	High
Weather data	WeatherXM	~$0.01/API call	Tomorrow.io	$0.05-0.20/call	10x	Medium
VPN relay	Anyone	Free relay (earn)	NordVPN	$3.49-8.32/mo	N/A	High
Community WiFi	Wayru/WiFiDabba	~$2-5/mo	ISP retail	$15-30/mo	6.4x	Med-High
Wearable health	CUDIS	~$69 ring	Oura Ring	$299 + $6/mo	4.3x	High
IoT infra (L1)	IoTeX	~$0.01/tx	AWS IoT Core	$1/M msgs	N/A	Medium

Table 5. DePIN vs. Web2 unit economics across 13 service domains.

5.5 Governance Concentration

Governance concentration across DePIN and DeFi protocols is analyzed in a companion cross-sectional study using a Nansen-grounded evidence-traced exclusion methodology across 52 protocols [34]. That study documents that holding-based concentration is higher in DePIN than DeFi after protocol-controlled address exclusion (under the headline voter-inclusive staking pass-through treatment, Mann-Whitney $p = 0.028$, Cohen's $d = 0.65$, a medium effect; a uniform staking-aggregation exclusion gives Cohen's $d = 0.75$ at $p = 0.018$ as a robustness check), robust across leave-one-out iterations and alternative exclusion specifications. Delegation amplifies governance concentration above holding-based levels in thirteen of eighteen protocols with sufficient governance data (ratios 2.5x to 25.6x), with five design-driven dispersion exceptions (ENS, GMX, HNT, JUP, LPT); infrastructure protocols Optimism (3.6x) and Arbitrum (3.1x) sit within the amplification pattern. The present paper uses protocol-specific governance observations as inputs to the design playbook below. The present paper uses protocol-specific governance observations as inputs to the design playbook below.

6. DESIGN PLAYBOOK

Every design recommendation below ties to an empirical or architectural anchor from Section 5; recommendations without such grounding appear in Supplementary Material S1 (Practitioner Brief).

6.1 Design Implications

The data reveal a sector where most protocols are simultaneously governance-concentrated and subsidy-dependent. Only four protocols show confirmed burn activity above zero: Helium (S2R 1.84), DIMO (4.24), GEODNET (0.219), and Hivemapper (0.183); Helium and DIMO have crossed fiscal parity (S2R ≥ 1.0), with GEODNET on a single-peaked absorption trajectory that ramped to a peak in late 2025 before easing (Table 7). The following implications derive from this paper's own empirical evidence.

Demand concentration as independent health metric. The "Who Burns?" analysis (Section 5.1.2) reveals that high S2R can coexist with concentrated demand, creating single-buyer fragility. Burns should be coupled to real service usage, not financial speculation; and demand concentration should be monitored alongside S2R as a separate health metric.

Burn architecture determines whether demand concentration is observable. Helium's direct-burn model, in which each Data Credit purchaser signs their own HNT burn, exposes demand concentration on-chain and reveals it to be high

(Section 5.1.2). Buy-and-burn and protocol-mediated models (GEODNET, DIMO) instead route burns through a foundation or treasury, which relocates the demand base to off-chain subscriptions and licenses and leaves customer concentration unobservable on-chain. Where service pricing must be predictable for enterprise buyers, a dual-token or credit-denominated model reduces reflexivity risk, but designers should note that such models also remove demand concentration from on-chain view.

Spatial staking as frontier coverage mechanism. GEODNET's SuperHex system (Section 5.2.5) ties staking requirements to geographic coverage gaps, creating an economic incentive for frontier deployment without protocol-managed hardware subsidies. This represents a design primitive unavailable to non-geographic protocols.

Unit economics subsidy decomposition. The DePIN vs. Web2 comparison (Table 5) shows pricing 1.4 to 80 times below incumbents, but three of five domains carry medium or high subsidy risk. Service verification remains the hardest design problem: token incentives must reward actual service delivery, not merely asset deployment, as the Helium HIP-147 transition demonstrates (Section 5.1.1).

Subsidy taper creates fiscal cliffs. DePIN protocols face domain-specific taper risks as token emissions decline. Subsidy taper scenarios for each domain are developed in Appendix C.

Service incentives must be verifiable. The Helium HIP-147 transition from coverage-based to service-weighted rewards illustrates the core DePIN design challenge. The S2R trajectory (Table 3) shows this shift's fiscal impact: burn-to-issuance ratios rose from 0.01 to above 1.0 as rewards tracked real data transfer.

Forking asymmetry strengthens voice. DePIN's physical assets create forking asymmetry: software can be forked freely, but hotspots and storage nodes cannot easily switch networks. This makes contestation infrastructure (appeals, mediation, constitutional protections) more important than in pure software protocols.

6.2 The Subtraction Test [Proposal]

Deploying all recommended mechanisms simultaneously would reproduce the premature complexity anti-pattern identified in Section 6.4. The design implications above map institutional conditions to mechanisms, but the following filter operationalizes a principle of mechanism parsimony: before adding any governance or tokenomic mechanism, the designer should answer three questions.

(1) What behavior does this mechanism create? State the intended behavioral change in one sentence. If it cannot be stated simply, the mechanism may not be understood by participants, violating the transparency principle. Example: "Time-locked staking reduces reflexive selling during price drops."

(2) What misunderstanding will this mechanism create? Identify the most likely misinterpretation. Every mechanism creates a mental model in participants' minds; if the intended model and the received model diverge, the mechanism will produce unintended behavior. Example: participants may interpret staking locks as illiquidity risk instead of commitment signaling, triggering the opposite of the intended effect.

(3) What is the smallest version that demonstrates the behavior without requiring belief? Strip the mechanism to its testable core. If the mechanism requires participants to trust that future conditions will materialize (e.g., "burns will increase when adoption grows"), it is not yet testable and should be deferred until the prerequisite condition exists. The Helium trajectory illustrates this: HIP-147's shift to service-based rewards worked because network usage already existed; it would have failed in 2020 when the network had coverage but no demand.

This filter is the practical complement to the design implications. Section 6.1 identifies what to deploy under which conditions; the subtraction test says when to stop. Together they encode the core design philosophy of this paper: institutional legitimacy requires not only the right mechanisms but the right sequence, earning complexity through demonstrated need, adding mechanisms only after the prior layer is tested. For GeoDePIN protocols specifically, this means spatial reward routing should be introduced only after the network has sufficient coverage data to know where deployment creates value, not before.

6.3 DePIN Health Dashboard [Inferred]

Drawing on the empirical data assembled in Section 5, Table 6 proposes a set of 12-month health targets for DePIN protocols entering the growth phase. These are not empirically derived thresholds but normative benchmarks grounded in observed ranges across operating protocols. They are intended as a practitioner tool: a dashboard that protocol teams and governance participants can use to assess whether incentive structures are on track.

Demand concentration as a second diagnostic axis. The Helium "Who Burns?" analysis (Section 5.1) reveals that S2R and demand distribution are independent dimensions of protocol health. S2R measures fiscal sustainability (are burns offsetting issuance?); demand concentration measures counterparty resilience (is demand broad or captive?). The interaction produces four regimes: (1) High S2R /

Low demand concentration = ideal, where burns exceed issuance and demand is distributed; (2) High S2R / High demand concentration = “single-buyer fragility,” where the protocol is fiscally healthy but depends on a few large buyers whose departure would collapse the burn rate; (3) Low S2R / Low demand concentration = “diffuse but weak demand,” where many participants use the network but not enough to offset issuance; (4) Low S2R / High demand concentration = “fragile and captive,” the worst case.
Helium in H2 2025 sits in regime 2: S2R crossed parity (1.84) but the top-5 burn signers account for approximately 90% of total burns. Analysts should apply this framing to every GeoDePIN with a burn mechanism. We propose tracking Top-5 burn share alongside S2R in the health dashboard below.

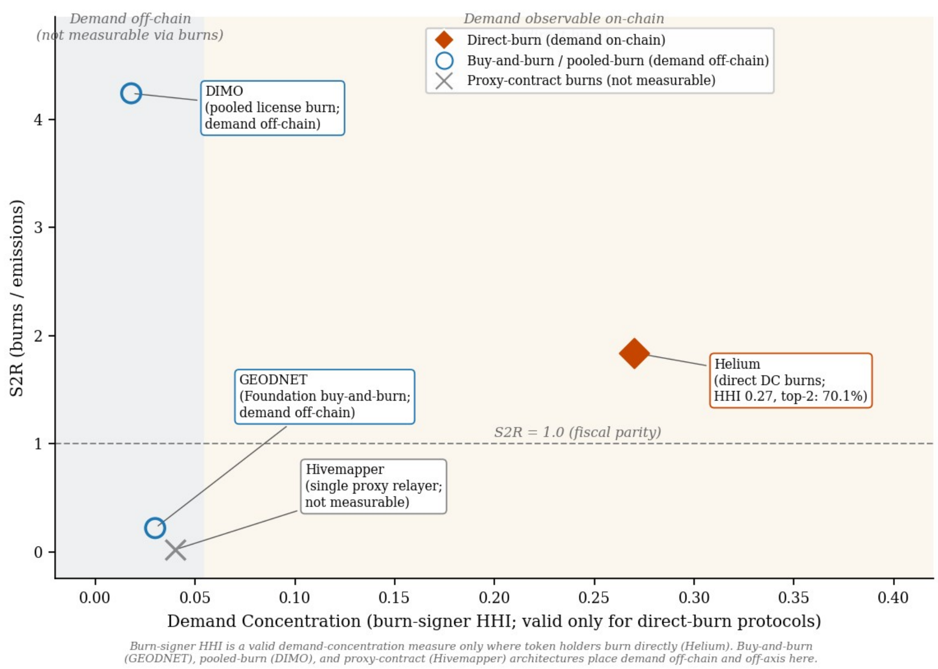

Figure 4. Burn-architecture observability and S2R. Burn-signer HHI is a valid demand-concentration measure only for direct-burn protocols, where token holders sign their own burns; among the protocols studied only Helium meets this condition (burn-signer HHI 0.27, S2R 1.84), and there demand is highly concentrated (single-buyer-fragility regime). Buy-and-burn (GEODNET) and protocol-mediated pooled-burn (DIMO) architectures route burns through a foundation or treasury, so their demand base is off-chain and is shown in the shaded off-axis lane at left rather than at a measured concentration. The horizontal line marks S2R = 1.0 (fiscal parity).

Three metrics anchor the dashboard (Table 6). S2R ≥ 0.35 means at least one-third of emissions are offset by usage-linked burns or locks. Helium far exceeds this target (S2R 1.84 as of February 2026 [33]), driven by carrier revenue burns and post-

halving issuance reduction; the health dashboard should track whether S2R remains above 0.35 over rolling 6-month windows, not single months.
Token stickiness (second metric): ≥ 60% means the majority of reward recipients hold their tokens beyond the initial vesting period; this is aspirational given that early Helium hosts widely sold rewards during the 2021–2022 speculative period. Owner-user overlap ≥ 20% signals a meaningful core of participants who both provide and consume the service.
Demand concentration (third metric): If Top-5 burn share exceeds 80% for 6 consecutive months, activate counterparty risk mitigations (new enterprise buyers, retail usage bundles, corridor-level incentives). This trigger converts the "Who Burns?" caveat into a general governance control.

The "Who Burns?" diagnostic measures demand concentration only where token holders burn directly. Among the protocols studied, only Helium meets this condition, and there demand is highly concentrated (burn-signer HHI 0.27, with roughly five purchasers accounting for about 90% of burns). Buy-and-burn architectures (GEODNET) and protocol-mediated pooled-burn architectures (DIMO) route burns through a foundation or treasury rather than customers, relocating demand to off-chain fiat subscriptions and licenses; for these architectures on-chain burn-signer concentration is not a valid proxy for customer concentration, and the underlying demand distribution is not observable on-chain. The diagnostic's applicability boundary is itself a finding: aggregate fiscal metrics and on-chain burns both fail to reveal the customer base of buy-and-burn and pooled-burn protocols.

The demand concentration diagnostic draws on a well-documented risk in corporate finance and supply chain management. Resource Dependence Theory [43] predicts that asymmetric reliance on a small number of buyers grants those buyers disproportionate influence over the platform's strategic direction, a structural vulnerability that empirical studies confirm increases cost of equity capital and induces strategic short-termism [44]. Helium's burn-signer HHI of 0.27, with two addresses controlling 70% of Data Credit burns, falls within the range this literature identifies as introducing material counterparty risk. For buy-and-burn and license-based protocols such as GEODNET and DIMO, the analogous demand base is off-chain (fiat subscriptions and developer licenses) and is not measurable from burn-signer data, so no comparable on-chain concentration figure can be reported; whether their revenue diversification reduces buyer leverage as Resource Dependence Theory predicts is a question this on-chain method cannot answer. The Top-5 threshold of 80% proposed above translates this corporate finance insight into a protocol-specific monitoring trigger. The demand concentration diagnostic also captures an oligopsony risk identified in industrial organization theory: when a small number of buyers account for the majority of network revenue, their bargaining power compresses operator margins and constrains protocol governance, a dynamic that is visible on-chain only for direct-burn protocols and that remains unmeasured for buy-and-burn and license-based models.

Metric	Target	Helium Actual (Feb 2026)	Basis
S2R (burns only)	≥ 0.35	1.84 (far exceeds)	>1/3 of emissions offset by usage sinks
Token stickiness	≥ 60% at 4wk	Unknown (improving)	Standard retention benchmark
Owner-user overlap	≥ 20%	Unknown (proxy only)	Minimum alignment signal
p95 node uptime	≥ 95%	~90% (dense), ~75% (sparse)	Infrastructure reliability floor
Post-incentive retention	≥ 60% at 4wk	Unknown	Anti-mercenary threshold

Table 6. DePIN health dashboard metrics with proposed 12-month targets, Helium actuals (February 2026), and empirical basis.

Protocol	Sink Type	S2R (burns)	S2R (burns+locks)	Period	Source	Demand HHI
Helium	Full DC burn	0.02-1.84	higher	Apr 2023–Feb 2026	Dune (on-chain)	0.27
Filecoin	Base fee burn + collateral	~0.045	0.14-0.41	Jun 2023	Deep Research	—
Livepeer	None (fees in ETH)	~0	~0	Q1 2025	DeFi Llama	0.31
Hivemapper	Map credit burn	0.183	0.183	Feb 2026	Dune Analytics (Feb 2026); burn-and-mint active but not queryable via Dune	—
GEODNET	Subscription burn	0.219	0.219	Feb 2026	Dune (Polygon buy-and-burn; Polygon net issuance)	—
DIMO	License burn	4.24	—	Feb 2026	Dune Polygon (on-chain)	—
IoTeX	Gas fee burn	N/A	N/A	Feb 2026	IoTeX L1 not indexed by Dune; gas fee burn active on native chain	—
WeatherXM	Subscription burn	N/A	N/A	Feb 2026	Subscription burn active; volumes too small for reliable S2R computation	—
Grass	None (staking only)	~0	~0	Feb 2026	Public docs	—
Anyone	Stake/lock	~0	~0	Feb 2026	Public docs	—

Table 7. Cross-protocol S2R and demand concentration summary across the 13-protocol sample. Column 'S2R (burns)' uses the burns-only definition; 'S2R (burns+locks)' includes token lock-up mechanisms where applicable. Demand HHI computed where per-address burn data is available. A long dash in the Demand HHI column marks protocols for which a valid on-chain demand-concentration measure is not available, either because per-address burn data is not queryable or because the protocol executes burns through a foundation or treasury (buy-and-burn and pooled-burn architectures), so on-chain burn signers are not customers.

Table 7 consolidates the S2R and demand concentration readings from Section 5 across the 13-protocol sample.

6.4 Risks and Trade-offs

Genuine trade-offs pervade every design choice above. No institutional design can simultaneously maximize all normative values; the art is in calibrating the balance for specific contexts.

Efficiency vs. Inclusivity. Adaptive efficiency (automated or expert-driven decisions) conflicts with deliberative inclusion (slower, participatory process). We saw this in Ethereum's EIP-1559 [5] fee mechanism, where automated base fee adjustment coexists with governance deliberation over parameter changes. The resolution is domain-specific: categorize decisions by urgency and reversibility, applying fast-track processes for routine parameters and full deliberation for constitutional changes.

Decentralization vs. Coherence. Polycentric governance [14] can tip into fragmentation. Helium's subnetwork architecture shows the mitigation: IoT and Mobile networks share a common governance framework but maintain separate reward pools, achieving local autonomy within coordinated boundaries. The constraint is that some domains demand uniform standards (security, interoperability), limiting how far polycentricity can extend.

Transparency vs. Privacy/Security. Transparency requirements clash with operational security and personal privacy. Full on-chain transparency of governance votes enables accountability but also enables vote-buying and coercion. The resolution is contextual: publish governance rules and aggregate outcomes publicly while protecting individual vote privacy where coercion risk is high.

Innovation vs. Stability. Experimentation is essential for institutional learning but risky for systems managing real value. The health dashboard (Section 6.3) addresses this by defining trigger conditions: innovate freely above safety thresholds, but invoke conservative governance when metrics breach critical levels.

Token Incentives vs. Intrinsic Motivation. Tokenomics can crowd out intrinsic motivation, as the Uniswap airdrop case illustrates (90.8% claim rate but only 7% long-term retention [8], [24]). The design implication is to use tokens for coordination and accountability, with reputation and recognition as complementary motivational levers, and to invest in reputation systems and community recognition alongside financial incentives.

Scale Limitations. Many mechanisms work in small communities; as protocols scale, direct participation becomes impractical and delegation concentration increases. The mitigation is layered governance: local autonomy for routine decisions, representative governance for system-wide issues, and constitutional protections at all levels.

Premature Mechanism Complexity. S2R data reveals an anti-pattern: protocols that deploy elaborate tokenomics before establishing demand-coupled burns show S2R near zero (Grass, UpRock, early Helium pre-2024), while protocols that earned complexity through iteration achieve meaningful S2R (Helium 1.84, DIMO 4.24, GEODNET 0.219). A mechanism too complex for ordinary participants to verify violates the transparency principle regardless of whether the code is open-source. The implication is that mechanism designers should treat complexity as a cost, not a feature, and deploy new mechanisms only after the prior layer has been tested and understood by participants.

7. CONCLUSION

7.1 Summary

The S2R data across 13 protocols reveal a three-tier sustainability hierarchy for DePIN: Helium (1.84) and DIMO (4.24) have crossed fiscal parity through demand-coupled burns; GEODNET (0.219) and Hivemapper (0.183) occupy a growth-phase middle tier with confirmed burn activity but continued subsidy dependence; and the remaining protocols cluster near S2R zero, the sector default. The Helium longitudinal case (Table 3) documents the trajectory: demand-coupled burns can drive S2R above fiscal parity when paired with service-based reward transitions (HIP-147), but demand concentration (top-5 signers at approximately 90% of burns) creates a shadow risk invisible to aggregate metrics. Governance concentration in GeoDePIN tokens (HHI 0.034 to 0.593) materially exceeds the post-exclusion DeFi benchmarks from a companion cross-sectional study (0.005 to 0.065), consistent with the 13-protocol cross-section. The four-loop design architecture (economic, operational, integrity, governance), grounded in institutional design principles, and the GeoDePIN vs. DeVIN taxonomy (Table 2) organize these findings into testable structure.

7.2 Contributions

GeoDePIN protocols face institutional design pressures that generic DePIN analysis obscures: location-dependent verification, hardware lock-in that raises exit costs, and subsidy cliffs that force fiscal sustainability. Four artifacts address these pressures: (1) a GeoDePIN vs. DeVIN taxonomy with testable predictions about governance concentration, verification complexity, and subsidy sensitivity; (2) a demand-concentration diagnostic (the "Who Burns?" analysis and the S2R x demand-concentration 2x2 framework), applicable to direct-burn protocols and demonstrated on Helium; (3) a design playbook keyed to observable protocol conditions with a subtraction test for minimum viable mechanism complexity; and (4) a health dashboard with demand-concentration guardrails (Top-5 > 80% AND Top-1 > 35% for 6 months triggers diversification action).

7.3 Limitations

The governance concentration cross-section is a February 2026 snapshot; concentration trajectories over time would strengthen causal claims. Unit economics use posted prices (N = 1 per service domain); we estimate subsidy decomposition. The Helium S2R trajectory is observational: HIP-147 was not randomized, and confounders cannot be fully separated. Several protocols remain data-insufficient; their metrics are unmeasured, not absent. The sample includes only protocols that have survived to 2026; failed or abandoned projects may exhibit different patterns. Design recommendations generalize from Helium (the primary longitudinal case) to other GeoDePIN protocols by analogy; direct replication across multiple protocols would strengthen external validity.

The S2R metric as currently specified measures burns against mining emissions; it does not capture vesting-driven dilution from team, investor, and ecosystem unlocks, which for GEODNET represented 54% of total supply by Q2 2025. A comprehensive fiscal sustainability metric would incorporate all supply sources; this extension is specified as a future research direction.

S2R measures fiscal sustainability and demand coupling but does not capture participant-level welfare: an S2R above 1.0 can coexist with operator financial distress when token-to-fiat conversion factors are adverse [48]. The San Jose pilot [50] demonstrates this decoupling during the pre-parity regime. Participant-level return-on-investment analysis (matching on-chain reward data with off-chain hardware costs, electricity consumption, and maintenance labor) remains a measurement priority.

More broadly, DePIN protocols claim to deliver public infrastructure goods (cheaper wireless, broader positioning coverage, fresher mapping data), yet every evaluation metric in the present study and in the existing literature measures token economy health rather than whether participants' quality of life actually improved. This welfare measurement gap is a structural property of on-chain data architectures, not merely a data limitation of the present study: protocols instrument token flows but not service outcomes. Connecting fiscal sustainability to participant welfare requires instrumentation that does not yet exist in the DePIN ecosystem.

7.4 Future Work

Three additional research directions emerge from the empirical findings. HIP-147 constitutes a named, dated governance action suitable for formal event study analysis, with strong pre/post data for S2R and weaker data for service quality outcomes. Simulation evidence [29] suggests a Slashing-to-Burn Ratio as a third diagnostic axis; empirical validation across operating protocols is needed.

Agent-based simulation of DePIN market dynamics using large language models (Liu and Schweitzer, 2025) offers a complementary methodology for stress-testing the health dashboard thresholds proposed here, particularly for modeling how concentrated buyers respond to emission halvings.

Longitudinal analysis of token economic failure patterns identifies tokenomics-market fit disconnect as a primary failure mode (Belkin, 2026), validating the health dashboard's emphasis on trajectory monitoring over point-in-time assessment.

Beyond these, two extensions have highest priority. First, longitudinal governance tracking: monthly HHI/Gini snapshots would convert the cross-section into a panel and enable deconcentration analysis. Second, capability measurement: most protocols do not instrument real human outcomes; partnering with protocols to collect participant-level data would fill the most important gap. The burn-concentration finding is asymmetric across burn architectures: Helium's burns are signed directly by Data Credit purchasers, so its top-5 burn signers (approximately 90% of burns) measure genuine customer concentration. Buy-and-burn protocols such as GEODNET collect subscription revenue off-chain in fiat and execute burns through a foundation treasury, so on-chain burn signers do not identify customers and burn-signer concentration cannot be measured for them. This asymmetry opens a research question that on-chain data alone cannot answer: whether subscription-based demand architectures systematically produce more distributed customer bases than carrier-contract models. Testing it would require protocols to disclose off-chain subscription and license concentration, since the "Who Burns?" diagnostic is valid only where token holders burn directly.

Resource Dependence Theory's causal predictions generate a forward hypothesis for the only protocol whose demand concentration is directly observable on-chain: because Helium's Data Credit purchasers burn HNT directly, its measured burn-signer HHI (0.27, top-2 addresses at 70.1% of burns) reflects real buyer concentration, and the theory predicts that such concentration raises the probability of simultaneous buyer exit and the associated S2R volatility. For protocols whose burns are executed by a foundation or protocol treasury rather than by customers, on-chain burn-signer HHI is not a valid measure of demand concentration, so the comparable test must draw on off-chain customer-base disclosure that does not yet exist. Longitudinal tracking of both S2R and demand HHI across halving events would provide the first empirical test of whether burn-model architecture determines fiscal resilience. Shapley value analysis of demand concentration would quantify whether the top burn addresses' probability of being pivotal in fiscal sustainability exceeds their nominal burn share. Helium's S2R = 1.0 threshold defines a binary characteristic function: coalitions of burn addresses whose combined contribution meets emissions earn value 1, others earn 0. The top-2 addresses (70.1% of burns) are not merely large contributors; they are the margin between self-sustaining and subsidy-dependent regimes, a distinction HHI captures in magnitude but not in pivotality.

Appendix A: Practitioner Brief

The Practitioner Brief translates the design architecture and playbook into an actionable checklist for token designers and protocol founders, including a token justification litmus test, the four-loop institutional design system, DePIN-specific reward decomposition guidance, reflexivity-aware fiscal policy rules, and a KPI dashboard specification. Available as Supplementary Material S1.

Appendix B: Policy Brief

The Policy Brief provides a governance and consumer-protection framework for regulators engaging with tokenized institutions, including a proposed Tokenized Institution Disclosure Template, oversight metrics (economic stability, governance legitimacy, service quality, privacy), safe-harbor criteria tied to governance quality, and a pragmatic implementation pathway. Available as Supplementary Material S2.

Appendix C: Subsidy Taper Scenarios

A central question for DePIN sustainability is what happens to service pricing when token emissions decline. The unit economics in Section 5.4 show DePIN pricing 1.4–80x below Web2 incumbents, but three of five domains carry Medium or High subsidy risk. The following qualitative scenarios assess each domain's resilience to a 50% emission reduction:

Filecoin (Low-Medium risk): Miners already bid competitively for storage deals and earn fees directly from clients. A 50% emission reduction would increase the FIL cost of storage per GB-month, but the collateral requirement already filters for economically viable operators. Storage pricing would likely rise but remain below AWS S3 given Filecoin's lower infrastructure overhead.

Livepeer (Medium risk): Orchestrators earn ETH fees directly from clients ($3,071/day average, February 2026). LPT staking rewards supplement income but are not the primary revenue source. A 50% LPT emission reduction would reduce staking yields, causing some orchestrators to exit, but the direct fee revenue model is structurally sound.

Helium Mobile (High risk): Hotspot operators are primarily compensated through HNT rewards, not subscriber fees. The $30/month unlimited plan is priced well below T-Mobile's $70, but the gap is funded by token emissions. A 50% emission reduction would require either raising the plan price, increasing subscriber density per hotspot, or relying on operators who are also subscribers (owner-user overlap). This is the domain where sustainability is most uncertain and where the S2R metric is most critical as an early warning indicator.

Hivemapper (Medium risk): On-chain burn data confirm active demand (S2R 0.183, February 2026), but burns offset less than one-fifth of issuance, leaving the mapping network in its growth phase. Sustainability depends on achieving sufficient commercial demand for fresh street-level imagery to drive S2R toward parity as token rewards taper.

These scenarios do not constitute predictions but rather inform the monitoring cadence recommended above: protocols with High subsidy risk should track S2R and stickiness monthly, while those with Low risk can review quarterly.

Biography

Zach Zukowski is a researcher studying the intersection of institutional design, political philosophy, and decentralized systems. His work applies normative frameworks from political economy and constitutional theory to the design and governance of tokenized infrastructure networks. His research focuses on how token incentive systems function as institutional architectures encoding rights, duties, and enforcement mechanisms analogous to public policy instruments.

Declaration of Competing Interest

The author served as Senior Investment Analyst at Borderless Capital through February 2026 and retains no decision-making role, carried interest, or position-dependent compensation with the firm. The author holds personal token positions in three protocols analyzed in this paper: HNT (Helium), DIMO, and GEOD (GEODNET); positions are small and were established prior to the research period. The research, analysis, and conclusions presented are the author's own; no sponsor, portfolio holding, or third party had editorial input or review rights prior to submission.

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CRediT Authorship Contribution Statement

Zach Zukowski: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization.

Data Availability

The data supporting the findings of this study are publicly available from on-chain sources (Dune Analytics, Helius DAS API, Blockscout) via the queries specified in Section 3.1. Processed datasets and replication scripts are archived at the public repository https://github.com/Research-Publications-and-Data/Tokenomics-As-Institutional_Design (tag v1.0.0-frontiers-submission for the companion paper [34]; B3 replication materials will be tagged at acceptance).

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